

Midterm Exam (Operating System)

答案卷請註明學號, 班別, 組別, 姓名, 以方便成績輸入. 滿分:100

1. F(T or F) (2%) Modern OS prefer preemptive CPU scheduling for its average waiting time is minimum. (It is the 'shorter job' to decrease to waiting time, not 'preemptive'.)
2. F(T or F) (2%) Disk Scheduler prefers I/O bound process.
3. F(T or F) (2%) Local variables are shared among sibling threads.
4. B (2%) Which one is more invariant (不變的)? (a) policy, (b) mechanism
5. (2%) Please state a scenario(情境) that makes a process switch its state from "waiting" to "ready". For example, parent process is in waiting state since it issued the wait() system call, then the child process exits.
6. (5%) System calls provide 'programming' interface to system services. What happened if an operating system provides 'only GUI' interface? (只能透過GUI發出system call, 有何後果?) 無法發展新的應用程式, 只能透過既有的GUI操作電腦.
7. (5%) (a) Please list the four evaluation (評估) methods for CPU scheduling. (b) Which one of them is fast but challenged by its answer only applied to input cases? (哪一種雖然快,但是只能用在特定的輸入)?
(a) deterministic modeling, queueing model, simulations, implementation, (b) deterministic modeling
8. (5%) (a) State the advantages and the disadvantages by using layered approach to develop an operating system. (b) What is its major difficulty in implementing modern operating system? (layered approach 的優缺點, 實作上面臨的困難) : Pros: simplicity of construction and debugging, Cons: Efficiency. Difficulty: appropriately defining the various layer.
9. (5%) Multilevel feedback queue是為了改良Multilevel queue所發明的演算法, 請問相比於Multilevel queue優點為何?且相比於Multilevel queue, 增加了哪些條件? 條件: process is allowed to move between queues, 優點: Flexibility, for example, I/O-bound process is easily identified and assigned high priority.
10. (5%) Please state the difference between SRT (Shortest Remaining Time) and preemptive-SJF (Shortest Job First). No difference
11. (5%) Google's Chrome browser opens each new website in a separate process. Would the same benefits (利益) have been achieved if instead Chrome had been designed to open each new website in a separate thread? Explain. No. The primary reason for opening each website in a separate process is that if a web application in one website crashes, only that renderer process is affected, and the browser process, as well as other renderer processes, are unaffected. Because multiple threads all belong to the same process, any thread that crashes would affect the entire process.
12. (10%) Open-source operating systems are available in source-code format, such as Linux. On the opposite, Closed-source operating systems, such as Microsoft Windows, provide binary code only. (a) Which side does Apple's Mac OS X belong to (OS X 屬於哪一邊)? Either or neither, the micro kernel of OS X is open sourced, but the operating system is closed sourced.(b)"If you can't beat them, join them"(打不贏就加入), follow this proverb (諺語), which side will you join to? explain. 中心思想是加入強隊, 問題是哪一邊比較強? Windows一般使用者較多, 但是Linux的開發陣容堅強. 各有擅場, 立場與觀點決定答案.
13. (10%) To write program in multicore system, there are five challenges(挑戰), such as testing and debugging. Please name two from other four, and explain using Sudoku Validator as an example. (A Sudoku puzzle uses a 9 * 9 grid in which each column and row, as well as each of the nine 3 * 3 subgrids, must contains all of the digits 1, 2, 3... 9.)(剩餘四項挑戰,請舉出兩項並用數獨驗證器來說明) 除了testing and debugging之外, 還有的四項挑戰是(1)identifying tasks, 將任務拆解為較小的tasks, 對數獨驗證器而言這不難, 有27個tasks. (2)balance, 如何將27個tasks 均衡的分到4個cores, (3)data splitting, 將81個元素拆解到27個task, 每一個task都要讀取相關的9個元素. (4)data dependency, 27個tasks之間沒有相依性, 但是還有一個main task, 需要整合27個tasks的結果
14. (20%) The following tasks are being scheduled using a preemptive, round-robin scheduling algorithm. Each task is assigned a numerical priority, with a higher number indicating a higher relative priority. In addition to the tasks listed below, the system also has an idle task (which consumes no CPU resources and is identified as P0). The idle task has priority 0 and is scheduled whenever the system has no other available task to run. The length of a time quantum is 10 units. If a task is preempted by a higher-priority task, the preempted task is placed at the end of the queue with a fresh time quantum. Please (a) draw the Gantt chart, and calculate (b) the CPU utilization (使用率) for the first 150 time units, and (b) the waiting time for each process.

Thread	Priority	Burst	Arrival
P1	20	20	0
P2	30	25	5
P3	20	25	15
P4	35	15	60
P5	10	30	100
P6	10	20	105

(0, p1) – (5, p2) – (30, p1) – (40, p3) – (50, p1) – (55, p3) – (60, p4) – (75, p3) – (85, p0) – (100, p5) – (110, p6) – (120, p5) – (130, p6) – (140, p5) – (150, end)
utilization = $1 - (15 / 150) = 90\%$

Thread	Burst	Arrival	End	Turnaround	Waiting
P1	20	0	55	55	35
P2	25	5	30	25	0
P3	25	15	85	70	45
P4	15	60	75	15	0
P5	30	100	150	50	20
P6	20	105	140	35	15

15. (20%) Explain the following terms. Please demonstrate your knowledge in associate with those terms, instead of translations. Examples are very welcome. Clear description helps credit.
(a)convoy effect (b)exponential averaging (c)loadable kernel module (d) processor affinity
解釋下列名詞。請展現對於作業系統的理解程度，切勿翻譯了事。盡可能多用例子。字數太少，資訊不足都很可能導致無法正確的評估。因此，請盡量多寫，寫清楚。惜墨如金的，責任自負。